Semester Project

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October 2026

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TODO: Citations, GUI, Future Work

1 SMC EX260-SDN1 DeviceNet

1.1 Physical Switch Settings

1.1.1 Communication Speed

The communication speed is set with the physical switches No. 3 and 4. The default value is 125kbps. [1, page 16]

Bitrate	Switch No.	
kbps	3	4
125	0	0
250	0	1
500	1	0

Table 1: Bitrates from Switch Settings

1.1.2 DeviceNet Address

The DeviceNet address is set with the physical switches No. 5 to 10. The default address is 63. [1, page 16]

Address	Switch No.					
-	5	6	7	8	9	10
0	0	0	0	0	0	0
1	0	0	0	0	0	1
2	0	0	0	0	1	0
...				...		
62	1	1	1	1	1	0
63	1	1	1	1	1	1

Table 2: Addresses from Switch Settings

1.2 Messaging Protocol

A message consists of a connection ID and a data field. Wait $\geq 1s$ for a response when you send a message. [2, chapter 2, page 11]

1.2.1 Connection ID (CID)

The CID counts 11 bits. The structure depends on the message group. Only group 2 is relevant for this project. The CID for group 2 contains the group ID (2 bits), the MAC ID (6 bits) and the message ID (3 bits). [2, chapter 2, pages 6-7]

	Group ID		MAC ID						Message ID		
Bit No.	10	9	8	7	6	5	4	3	2	1	0

Table 3: Group 2 CID Structure

Message ID	Function
4	Master's Explicit Request
6	Master/Slave Connection Management
7	Duplicate MAC ID Check

Table 4: Relevant Group 2 Message IDs

1.2.2 Data Field

The data field may include some flags. The R flag indicates a request or response. The F flag indicates a complete or fragmented message. The X flag is a transaction ID that's echoed in a response to match a request. [2, chapter 2, pages 22-24]

Flag	0	1
R	Request	Response
F	Complete	Fragmented
X	-	-

Table 5: Data Field Flags

1.3 Communication Process

The communication process involves checking for a unique MAC ID before allocating a master/slave connection. The connection times out after a few seconds. If no response is received for a master's explicit request, the master/slave connection requires reallocation.

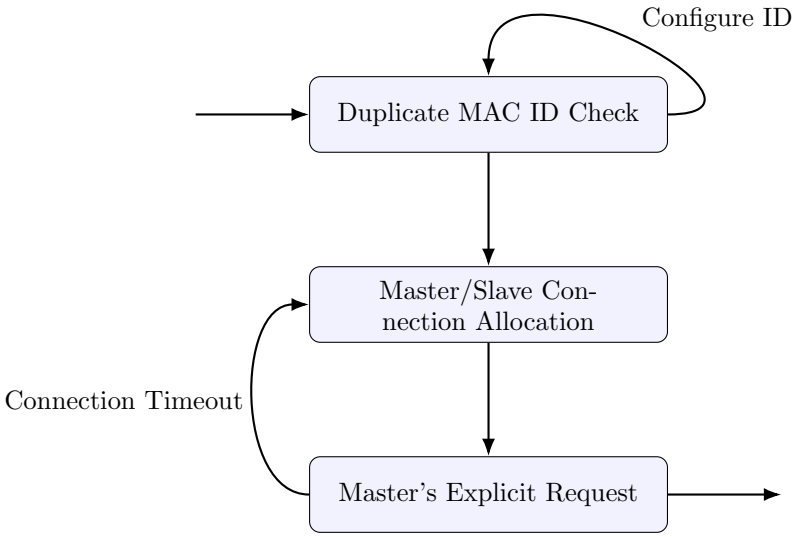


Figure 1: Communication Setup Process

1.3.1 Duplicate MAC ID Check

The message ID 7 is reserved for the duplicate MAC ID check in group 2. It requires human intervention to configure a unique MAC ID as it's inevitable that two modules on the same link will be assigned the same address. The duplicate MAC ID check has to be transmitted two consecutive times without receiving a subsequent request or response message before establishing a connection. [2, chapter 2, pages 11-14]

Group ID		MAC ID						Message ID		
1	0	0	0	0	0	0	0	1	1	1

Table 6: CID for Duplicate MAC ID Check

Byte No.		
0	R	Physical Port Number
1		Vendor ID
2		
3		Serial Number
4		
5		
6		

Table 7: Data Field for Duplicate MAC ID Check

1.3.2 Master/Slave Connection Management

The message ID 6 is reserved for the master/slave connection management in group 2. The service code 0x4B indicates the allocation of a master/slave connection and the service code 0x4C indicates the release. The class ID 3 directs the message at the DeviceNet object. There exists exactly one DeviceNet object per physical attachement to the DeviceNet. Therefore, the instance ID is always 1. The allocation choice byte denotes an explicit message and I/O connections from the predefined master/slave connection set. The acknowledge suppression bit has to be 0 for the release. [2, chapter 3, pages 13-24]

Group ID		DeviceNet Address						Message ID		
1	0	1	1	1	1	1	1	1	1	0

Table 8: CID for Master/Slave Connection Management

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x4B]	
2	Class ID [0x03]		
3	Instance ID [0x01]		
4	Allocation Choice		
5	DeviceNet Address		

Table 9: Data Field for Master/Slave Connection Allocation

Byte No.		
0	F	X MAC ID
1	R	Service Code [0x4C]
2		Class ID [0x03]
3		Instance ID [0x01]
4		Release Choice

Table 10: Data Field for Master/Slave Connection Release

Bit No.	Connection
0	Explicit Message
1	Polled
2	Bit Strobed
3	Multicast Polling
4	Change of State
5	Cyclic
6	Acknowledge Suppression
7	Reserved

Table 11: Allocation/Release Choices Byte

1.3.3 Master's Explicit Request

The message ID 4 is reserved for the master's explicit request in group 2. The service code 0x10 indicates a write to a single attribute and the code 0x0E a read. [2, chapter 2, pages 36-37] The class ID 9 represents the discrete output points. The class ID 4 represents the Assembly Object. The instance ID 37 corresponds to the 32 solenoid output points. [1, pages 20-22] The 4 value bytes act as a mask to set all 32 channels at once. The instance ID corresponds to the desired channel.

Group 2		DeviceNet Address						Message 4		
1	0	1	1	1	1	1	1	0	1	0

Table 12: CID for Master's Explicit Request

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		

Table 13: Data Field for Master's Explicit Read

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x09]		
3	Instance ID		
4	Attribute ID [0x03]		
5	Value		

Table 14: Data Field for Master's Explicit Write

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x0E]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		

Table 15: Data Field for Master's Explicit Read All

Byte No.			
0	F	X	MAC ID
1	R	Service Code [0x10]	
2	Class ID [0x04]		
3	Instance ID [0x25]		
4	Attribute ID [0x03]		
5	Value		
6			
7			
8			

Table 16: Data Field for Master's Explicit Write All

2 Keller PAA-4LD I^{2}C

2.1 Request Measurements

The command to request a measurement is 0xAC. Wait $\geq 8ms$ before reading the measurement.

2.2 Read Measurements

The response contains the status (1 byte) and the measurement (2 bytes). The status carries a checksum error indication at bit 2. The two data bytes are stitched together and then scales with $(p - 16384) * (p_{\max} - p_{\min})/32768 + p_{\min}$ for the pressure. The absolute pressure ranges from $p_{\min} = 0$ bar to $p_{\max} = 3$ bar.

3 Autonics TM4-N2SB RS485

Code	Address (Channel i)	Description (Run=0, Stop=1)
1	0x0002 * i	Read Heating Status
5	0x0002 * i	Write Heating Status
3	0x03E8 * i	Read Set Temperature
6	0x03E8 * i	Preset Temperature
4	0x03E8 + 6 * i	Read Temperature

Table 17: Function Codes and Addresses for Autonics

4 Elemental Analyzer (EA) TCP/IP

The EA device communicates over the TCP/IP protocol. The messages are sent to the IP address of the EA device via a dedicated port.

Message	Description
STRT	Start device
SLEEPON	Send device to sleep
?STS	Get device status
SEQON	Start processing
SEQOFF	Halt processing
SEQ	Get process status
?SINX	Get sample index `i`
?NAM `i`	Get sample name
?WGH `i`	Get sample weight
?PCT `i` C	Get sample C%
?PCT `i` N	Get sample N%

Table 18: Messages for the EA Device

5 Graphitization

The graphitization process is split into 3 stages: Conditioning the device, loading the samples and the reaction. Each stage has their own procedures. Some procedures can be executed in parallel because they operate on separate sections of the device. The following steps describe the default graphitization process.

5.1 Conditioning

5.1.1 Oxidize Iron

1. Open the line valves
2. Heat the ovens to 500°C
3. Lift the ovens to the reactors
4. Wait 5mins for the irons to preheat

5.1.2 Clean Trap

5. Open the sampling valve for helium from the EA
6. Wait 1min for the trap to be flushed
7. Close the sampling valve

5.1.3 Leak Test

8. Open the reactor valves and the vacuum valve
9. Wait 3mins for the reactors to be evacuated
10. Close the reactor valves and the vacuum valve
11. Check the pressure difference over 5mins

5.1.4 H2 Cleaning

Repeat H2 Cleaning (12. - 15.) ?? times

12. Open the hydrogen valve
13. Wait until one of the reactors reaches 800mbar
14. Close the hydrogen valve
15. Wait 1min

5.1.5 Correct Offset

16. Open the reactor valves and the vacuum valve
17. Wait 20s for the reactors to be evacuated
18. Store the remaining pressure as offset
19. Close the reactor valves and the vacuum valve

5.2 Load Samples

Repeat Load Samples (20. - 54.) for every active reactor

20. Wait for the EA status "STS READY"
21. Store the sample index, name and weight
22. Open the vortex valve
23. Wait for the trap to reach 10°C
24. Close the vortex valve
25. If the sample name starts with "first" (any capitalization) go to Load Samples (20.)
26. If the sample name starts with "pre" (any capitalization) go to Clean Trap (52.)

5.2.1 Sampling

27. Start the EA
28. Wait for the EA status "RUN"
29. If the sample name starts with "runin" (any capitalization) go to Clean Trap (52.)
30. Start processing the sample
31. Wait for the processing status 10
32. Open the sampling valve
33. Wait for the processing status 15
34. Close the sampling valve
35. Stop processing the sample
36. Get the sample C% and N%

5.2.2 Load Reactor

37. Open the trap valve and the vacuum valve
38. Wait 3s for the trap to evacuate
39. Open the reactor valve
40. Wait 10s for the reactor to evacuate
41. Close the vacuum valve
42. Heat the trap to 420°C
43. Wait 10s for the sample to be loaded to the reactor
44. Close the reactor valve and trap valve
45. Store the pressure as CO2 pressure
46. Open and close the vacuum valve and the reactor valve alternately until the pressure reaches ??
47. Store the pressure as CO2 pressure final
48. Open the reactor valve and the hydrogen valve
49. Wait until the pressure reaches ??
50. Open and close the vacuum valve and the reactor valve alternately until the pressure reaches ??
51. Store the pressure as H2 pressure

5.2.3 Clean Trap

52. Open the sampling valve for helium from the EA
53. Wait 1min for the trap to be flushed
54. Close the sampling valve

5.3 Reaction

5.3.1 Reaction

55. Start the peltier coolers
56. Heat the ovens to 580°C
57. Wait 2h for the reaction
58. Lower the ovens from the reactors
59. Stop the ovens

5.3.2 Remove Water

60. Open the reactor valves and the vacuum valve
61. Wait 7mins for the reactors to be evacuated
62. Close the reactor valves, the line valves the vacuum valve
63. Stop the peltier coolers

6 Graphical User Interface (GUI)

6.1 Control

The control class is central to the GUI. It holds several other classes such as the devices, the procedures, the samples table, the pressure history plot as well as the schematic. The communication between these classes is connected by the control class.

6.2 Schematic

The schematic represents the graphitization device. It shows the reactors, heaters, coolers and valves. The reactors show the current pressure if they are active. They can be deactivated with a button click. If a line of reactors is deactivated, that side won't be opened, heated or cooled during the graphitization process. The heaters and coolers will display if they are running. The valves can be manually opened or closed at any time with a click.

6.3 Graph

The graph plots the active reactors pressure history.

6.4 Table

The table holds sample data. The user can enter the sample name and the values are populated during the graphitization process. The table is not yet connected.

6.5 Buttons and Displays

The GUI holds 3 buttons: Start, stop and reset. Clicking start will run the graphitization process. The stop button can be clicked at any time to stop the process, turn off the heaters and coolers and close all valves. With the reset button, the graph starts fresh and the pressure offset is cleared.

There is a display for every stage of the graphitization: Conditioning, load samples and reaction. These headers can be clicked to start only this section of the process. Underneath, the step that is currently running is displayed.

7 Threading

The GUI runs on the main thread. Further threads are introduced to keep the GUI responsive at all times. Every device has its own lock. Only one thread at a time can acquire the lock and communicate to the device. There are dedicated threads for every device that regularly poll new data. The procedures are executed in a separate thread that can request writes to the devices. The procedures emit different signals such as a start, end and prompt to interact with the GUI.

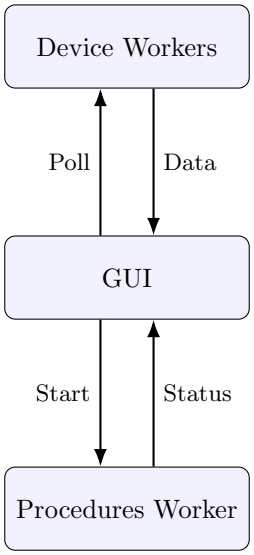


Figure 2: GUI and Worker Interactions

8 Configurations

8.1 SMC

DeviceNet Address	MAC ID	Bitrate
63	0	125000

Table 19: DeviceNet Configuration

Name	ID
Reactor 1	1
Reactor 2	2
Reactor 3	3
Reactor 4	4
Reactor 5	5
Reactor 6	15
Reactor 7	14
Reactor 8	13
Reactor 9	12
Reactor 10	11
Line 1	6
Line 2	10
Vacuum	7
Trap	8
H2	9
He	16
Vortex	17
-	18
Lift	19

Table 20: Valves Configuration

8.2 Keller

Reactor No.	Address
Reactor 1	0x50
Reactor 2	0x51
Reactor 3	0x52
Reactor 4	0x53
Reactor 5	0x54
Reactor 6	0x55
Reactor 7	0x56
Reactor 8	0x57
Reactor 9	0x58
Reactor 10	0x59

Table 21: Keller Configuration

8.3 Autonics

Port	Address	Baudrate
/dev/ttySC0	1	9600

Table 22: Instrument Configuration

Name	ID
Trap	1
Oven 1	2
Oven 2	3

Table 23: Channel Configuration

9 Future Work

9.1 Configurations Tab

9.2 Storing Data

References

[1] SMC Corporation, *SI Unit for DeviceNet Operation Manual: EX260 Series*, Jul. 2016, available at https://www.smcworld.com.

[2] ODVA, *The CIP Networks Library, Volume 3: DeviceNet Adaptation of CIP*, 1st ed., Nov. 2018, specification document.